Environmental Studies — Week 5

Climate, Carbon and Environmental Monitoring

Illustrated notes: mechanism first, then diagram, then worked example

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1 The greenhouse effect, derived rather than memorised

Definition

The **natural greenhouse effect** is the warming of Earth's surface caused by gases in the troposphere (water vapour, CO_2, CH_4, N_2O) that are nearly transparent to incoming sunlight but absorb outgoing infrared (heat) radiation and re-emit part of it back downwards. **Global warming** is the *human-enhanced* part of this warming; **climate change** is the long-term shift in temperature and weather patterns that results.

Fixing a common misconception

"A greenhouse gas layer works like the glass of a garden greenhouse." Only loosely. Glass warms a greenhouse mainly by stopping warm air from *leaving by convection*. The atmosphere does something different: it intercepts *infrared radiation*. (Your solar cooker in Week 6 is the closer analogue: glass lets visible light in and traps the infrared it becomes.)

"The greenhouse effect is the ozone hole." No. The ozone hole is stratospheric ozone loss (more UV reaches the ground); the greenhouse effect is about infrared absorption in the troposphere. They are separate problems with some overlapping chemicals.

"The greenhouse effect is bad." The *natural* effect is what makes Earth habitable; the problem is the *enhancement*.

1.1 Why Earth would be about $-18\,^\circ$C without it

WHY this is true

In steady state, energy in = energy out. Sunlight arrives on a disc of area πR^2 but the planet's surface area is $4\pi R^2$, so the *average* incoming flux is $S_0/4$. A fraction α (the albedo, about 0.3) is reflected. A surface at temperature T radiates σT^4 (Stefan–Boltzmann, $\sigma = 5.67 \times 10^{-8}$ W m^{-2}K^{-4}). Balance:

$$\frac{S_0}{4}(1-\alpha) = \sigma T_e^4 \quad \Rightarrow \quad T_e = \left[\frac{S_0(1-\alpha)}{4\sigma}\right]^{1/4}.$$

Worked example

$S_0 = 1361$ W/m^2, $\alpha = 0.3$.

$$\text{absorbed} = \tfrac{1361\times 0.7}{4} = 238 \text{ W/m}^2 \ (\approx 240)$$

$$T_e = \left(\tfrac{238.2}{5.67\times 10^{-8}}\right)^{1/4} = (4.20\times 10^9)^{1/4} = 254.6 \text{ K} \approx -18.6\,^\circ\text{C}.$$

The measured global mean is about 288 K (+15 °C). The ~ 33 K gap is the natural greenhouse effect.

1.2 One absorbing layer shows how the extra warmth arises

Now add a single atmospheric layer that is transparent to sunlight but *absorbs all* infrared and re-emits it equally up and down. Let the sun supply 240 units to the surface (and nothing is absorbed in the layer from sunlight).

WHY this is true

Top of atmosphere: the layer is the only thing radiating to space, so it must emit 240 upward. It emits equally up and down, so it also emits 240 downward.
Surface: it receives 240 (sun) + 240 (from the layer) = 480, so it must emit 480.
Layer check: it absorbs the surface's 480 and emits $240 + 240 = 480$ ✓.
So $\sigma T_s^4 = 480 = 2 \times 240 \Rightarrow T_s = 2^{1/4} T_e = 1.19 \times 254.6 \approx 303$ K.

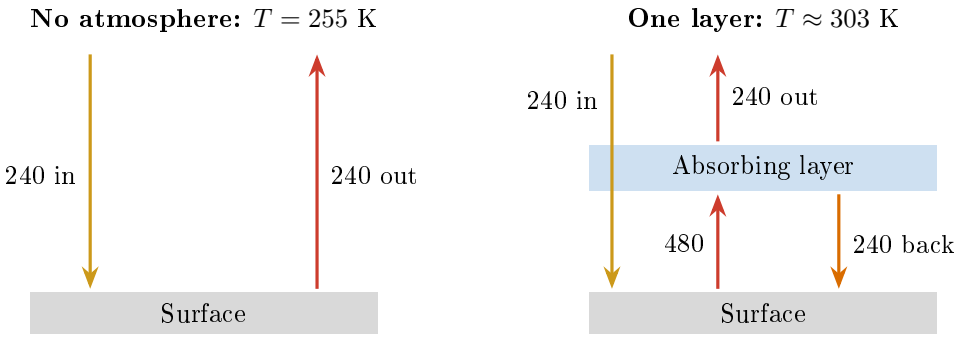


Fig. 1. Same sun, same top-of-atmosphere output (240), yet the surface carries double *the radiation. Notice the layer does not create energy: the surface is warmer because it now receives* 240 *from the sun* plus 240 *from the layer. Real air absorbs only part of the infrared, so reality (*288 *K) sits between* 255 *K and* 303 *K.*

1.3 Why sunlight passes in but heat is trapped: wavelength

WHY this is true

Hot bodies emit shorter wavelengths. Wien's law gives the peak: $\lambda_{\max} = 2898\ \mu\text{m K}/T$. Sun (5772 K): $\lambda_{\max} \approx 0.50\ \mu$m (visible). Earth (288 K): $\lambda_{\max} \approx 10\ \mu$m (infrared). CO_2 absorbs strongly near 15 μm and water vapour in several bands, i.e. *inside the Earth's* emission range but *outside the Sun's*. That mismatch is the entire mechanism.

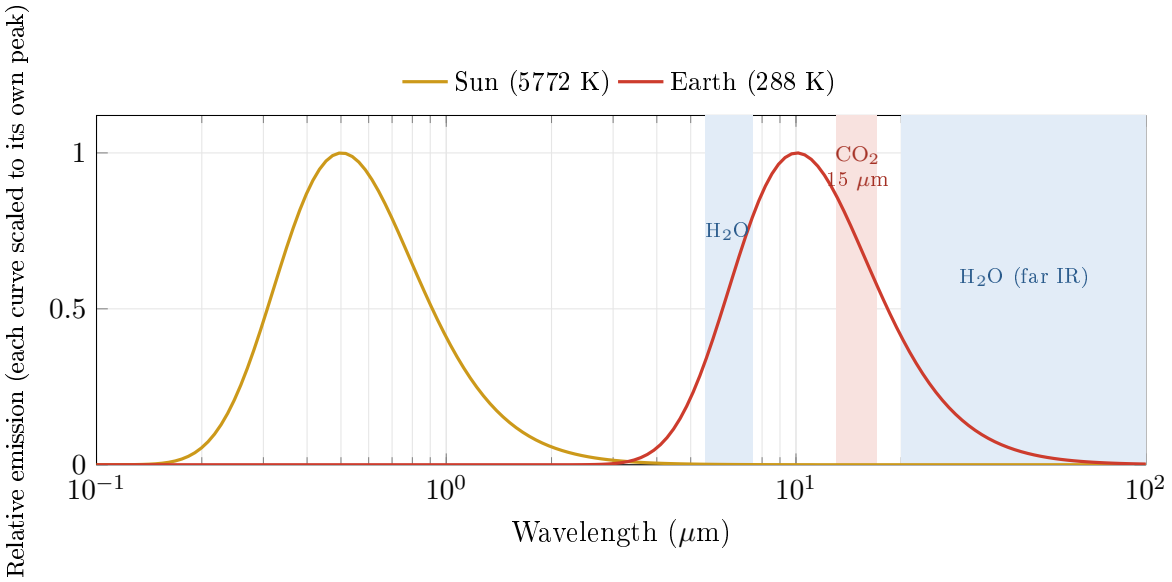


Fig. 2. The two curves barely overlap. Shaded bands are where greenhouse gases absorb: they sit under the red (Earth) curve and almost nowhere under the yellow (Sun) curve.

2 Not all greenhouse gases are equal

Definition

Global Warming Potential (GWP) compares the warming caused by 1 kg of a gas with 1 kg of CO_2, integrated over a fixed horizon (usually 100 years). CO_2 is defined as 1. **Lifetime** is how long a pulse of the gas stays in the air.

Gas	Main human sources	GWP_{100} (typical)	Behaviour
Carbon dioxide	fossil fuels, cement, deforestation	1	largest total contribution; very long-lived
Methane CH_4	livestock digestion, rice, landfills, leaks	≈ 28	more powerful per kg, but lifetime only about a decade
Nitrous oxide N_2O	nitrogen fertiliser, combustion	≈ 265	accumulates over decades–centuries
Fluorinated gases	refrigeration, equipment	up to $\sim$ 23,000 (course)	tiny quantities, extremely strong

The course states the "up to 23,000" figure for fluorinated gases; the ≈ 28 and ≈ 265 are typical IPCC-style values added here for the calculation below.

Worked example

A leaking pipeline releases 2 tonnes of CH_4. In CO_2-equivalent: $2 \times 28 = 56$ t CO_2e. This is why capturing biogas methane (Week 6) is a climate win even before counting the energy.

Fixing a common misconception

"Methane is the bigger problem because its GWP is bigger." GWP is *per kg*. Total warming = GWP $\times$ amount emitted $\times$ how long it persists. CO_2 dominates because humans emit it in enormous quantity and it lasts centuries; methane matters because it acts fast and is cuttable fast.

3 Natural versus human causes: attribution by elimination

WHY this is true

The course lists the natural drivers (reflectivity changes, orbit and tilt, solar output, volcanoes, natural CO_2 swings). The logic of attribution is to test each one against *two* questions: does it act on the right **timescale**, and does it point in the right **direction** for warming since the 1950s?

Natural driver	Timescale	Why it cannot explain post-1950 warming
Orbit and axial tilt (Milankovitch)	$\sim$ 20,000–100,000 yr	far too slow; it paces ice ages
Solar output	11-yr cycle	satellite record since 1978 shows no net rise while temperature rose
Volcanic aerosols (SO_2)	1–few yr	they *cool*, and settle out quickly
Natural CO_2 changes	thousands of yr	acted as a feedback amplifying orbital forcing, not as the initial cause
Surface reflectivity (ice, forests, cities)	decades	real, but mostly a *feedback* to warming (Section 4)

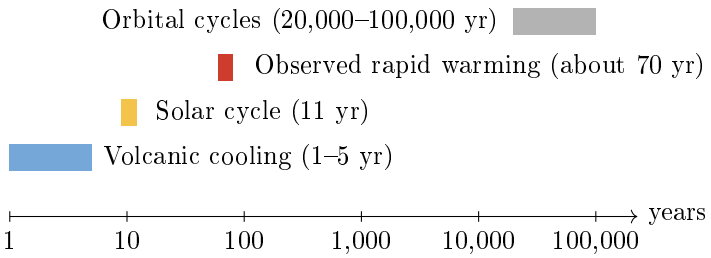


Fig. 3. A timeline on a logarithmic axis. The observed warming happened in about 70 years; orbital drivers need tens of thousands. Volcanic and solar effects are either brief or flat over the period.

Check against your course notes

The course says CO_2 is "52% higher than pre-industrial" (2020) and "414.8 ppm". The percentage depends on the baseline you pick: $414.8/280 \approx 1.48$ (48%), while a $\sim$ 272 ppm baseline gives $\approx$ 52%. For MCQs use the course figure; for understanding, know that the rise is roughly half again above pre-industrial, to levels not seen for millions of years. Also: the same notes give $>$ 95% certainty ("extremely likely") that humans are the dominant cause of post-1950 warming.

Human sources in the course list: burning coal, oil and gas (CO_2, N_2O); deforestation (lost sink *and* released stored carbon); manufacturing (cement, steel, plastics); livestock (methane); nitrogen fertiliser (N_2O); fluorinated gases.

4 Feedbacks: why small pushes become big changes

Definition

A **feedback** is a loop in which a change causes effects that change the original quantity again. A **positive** (reinforcing) feedback *amplifies* the initial change; a **negative** (balancing) feedback *reduces* it.

Fixing a common misconception

"Positive feedback means good feedback." In systems language, positive only means *same direction*: warming $\rightarrow$ more warming. Here every named feedback is positive, and that is exactly why they are worrying.

4.1 Ice–albedo feedback (worked out)

Worked example

Take incoming sunlight = 200 W/m^2 on a patch. Using the course figure, ice and snow reflect about 85% ($\alpha = 0.85$) while open ocean reflects about 7% ($\alpha = 0.07$).

$$\text{absorbed by ice} = 200 \times (1 - 0.85) = 30 \text{ W/m}^2$$
$$\text{absorbed by ocean} = 200 \times (1 - 0.07) = 186 \text{ W/m}^2$$
$$\text{ratio} = 186/30 \approx 6.2$$

Replace ice by water and that patch absorbs about six times more energy. That heat melts more ice, exposing more water.

4.2 Permafrost and glacial-cycle CO_2

Permafrost soils frozen for up to $\sim$ 40,000 years hold trapped carbon. Thawing lets microbes decompose it and release CO_2 and CH_4, which warm the planet and thaw more soil. Over glacial cycles, warm periods carried higher CO_2 and cold periods lower CO_2; this natural feedback amplified temperature swings that orbital changes started.

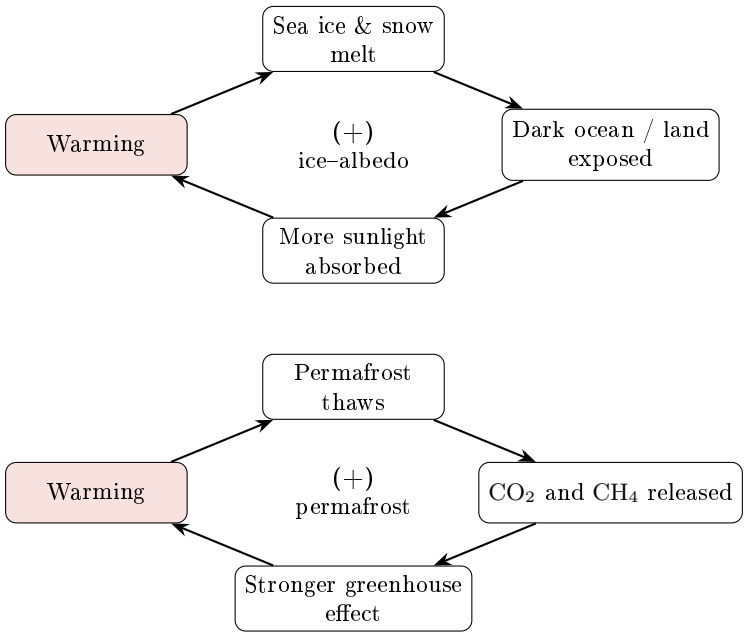


Fig. 4. Two reinforcing loops. Follow any arrow chain and you return to "Warming" with a bigger push than you started with.

5 The eight impacts, each with a mechanism

5.1 More frequent and severe weather

WHY this is true

Warmer air can hold more water vapour (Clausius–Clapeyron). Using the Magnus approximation for saturation vapour pressure, $e_s(T) = 6.112 \exp\left(\frac{17.62\,T}{243.12+T}\right)$ hPa with T in °C: $e_s(15) = 17.0$ hPa and $e_s(16) = 18.1$ hPa, so $+1\,°\text{C} \Rightarrow +6.6\%$ capacity. More moisture in the atmosphere means heavier downpours when it rains, while more evaporation dries soils

between rains, which explains *both* the floods and the droughts in the course notes.

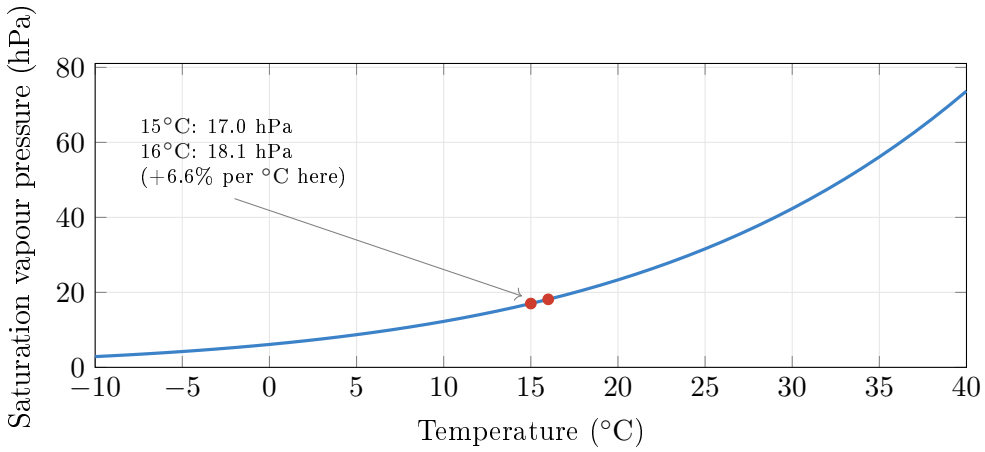


Fig. 5. The curve steepens with temperature, so each extra degree adds slightly more vapour than the last.

5.2 Melting ice caps and permafrost

Covered in Section 4: both are reinforcing feedbacks as well as impacts.

5.3 Higher air pollution: ground-level ozone

Fixing a common misconception

"Ozone is always good." Stratospheric ozone shields us from UV; **ground-level (tropospheric) ozone** is a pollutant and the main component of smog. It is not emitted directly: NO_2 is split by sunlight, $NO_2 + h\nu \rightarrow NO + O$, and then $O + O_2 \rightarrow O_3$, with volatile organic compounds sustaining the cycle. Heat and sunlight speed this up, so warmer weather means more smog, more asthma admissions, and more airborne pollen.

5.4 More acidic oceans

WHY this is true

Ocean water absorbs about a quarter of human CO_2. Dissolved CO_2 forms carbonic acid, which releases hydrogen ions: $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$. The extra H^+ combines with carbonate ions, $H^+ + CO_3^{2-} \rightarrow HCO_3^-$, leaving *less carbonate* for animals that build shells and skeletons from calcium carbonate (molluscs, crabs, corals).

Worked example

pH is logarithmic: $[H^+] = 10^{-\mathrm{pH}}$. A drop from pH 8.2 to 8.1 multiplies $[H^+]$ by $10^{0.1} = 1.26$, i.e. a **26% increase** from a change that looks tiny. Note the ocean is still alkaline (pH > 7): "acidification" means *moving toward* acidic, not becoming acidic.

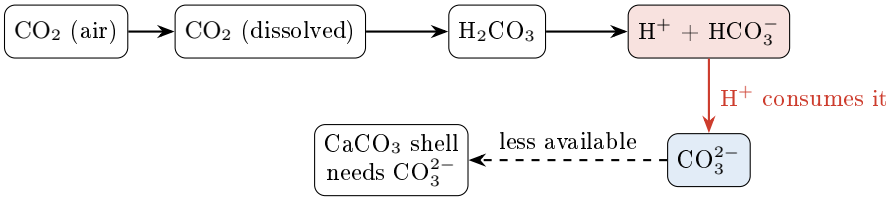


Fig. 6. The chemistry chain behind ocean acidification.

5.5 Rising sea levels

WHY this is true

Two contributions: (i) **thermal expansion**, since water expands when it warms; (ii) **land-ice melt**, which adds new water. For expansion alone, $\Delta h = \beta\, H\, \Delta T$.

Worked example

Warm the top 700 m of ocean by 1 °C with $\beta \approx 2.1 \times 10^{-4}\,\mathrm{K}^{-1}$ (illustrative): $\Delta h = 2.1 \times 10^{-4} \times 700 \times 1 = 0.147$ m ≈ 15 cm, with no melting at all.

Fixing a common misconception

"Melting Arctic sea ice raises sea level." Floating ice already displaces its own weight of water (Archimedes), so melting it changes the level almost not at all (a tiny effect arises only because fresh meltwater is slightly less dense than seawater). *Land* ice (Greenland, Antarctica, glaciers) is the dangerous one, because that water was not in the ocean before.

5.6 Higher death rates and wildlife extinction

Heat waves, pollution, floods and vector-borne disease raise illness and death, hitting children, the elderly and low-income communities hardest. For wildlife, species move toward cooler latitudes or altitudes and change migration timing; the course cites a 2015 study in which vertebrates are disappearing about 114× faster than the natural background rate, linked with climate change, pollution and deforestation.

Impact	Mechanism (one line)	Feedback to warming?
Extreme weather	warmer air holds $\sim 7\%/°\mathrm{C}$ more vapour	indirect
Melting ice	albedo falls, more absorption	**positive**
Permafrost thaw	CO_2 and CH_4 released	**positive**
Ground-level ozone	heat and sunlight speed smog chemistry	minor
Ocean acidification	$CO_2 \rightarrow H^+$, carbonate falls	weakens ocean sink
Sea-level rise	thermal expansion + land-ice melt	none directly
Death rates and extinction	heat, disease, habitat shift	none directly

6 Carbon sequestration: putting carbon back

Definition

Carbon sequestration is the long-term storage of CO_2 (or other carbon forms) outside the atmosphere to slow global warming. A **carbon sink** absorbs more carbon than it releases; a **source** releases more than it absorbs. The course groups methods as *biological*, *physical* and *chemical*.

WHY this is true

The natural carbon cycle swaps carbon among the atmosphere, biosphere, soil (pedosphere), rocks (geosphere) and oceans (hydrosphere). Burning fossil fuels moves carbon from the slow, deep reservoir (rocks) into the fast one (air) far quicker than natural processes move it back. Sequestration is deliberately speeding up the return trip. Always ask two questions of any method: **how much can it hold** and **how long does it stay put (permanence)**?

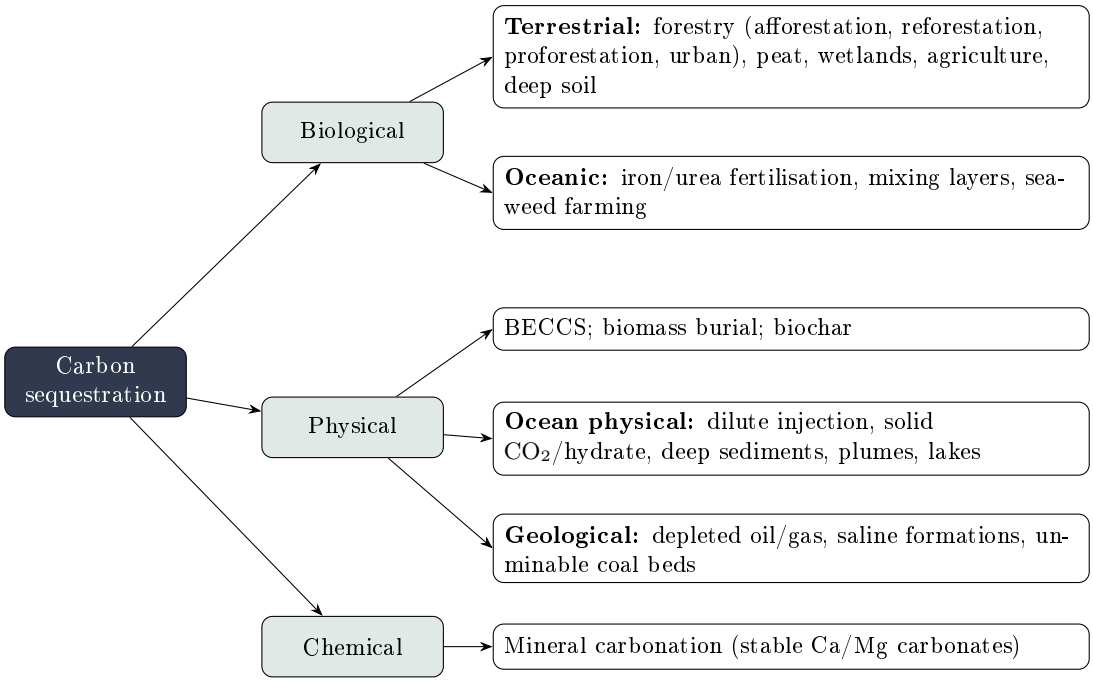


Fig. 7. The course's sequestration methods as a tree. Biological methods use life (photosynthesis); physical methods move or bury carbon; chemical methods lock it into a mineral.

7 Biological sequestration on land

7.1 One model explains peat, soil and cropland

WHY this is true

Let C be the carbon stored in a pool (a soil, a bog), I the yearly carbon input (plant litter, roots) and k the fractional decay rate per year (microbes turning it back into CO_2). Then

$$\frac{dC}{dt} = I - kC \;\Rightarrow\; C(t) = \frac{I}{k}\left(1 - e^{-kt}\right), \qquad C_{\text{eq}} = \frac{I}{k}, \quad \tau = \frac{1}{k}.$$

Two levers raise the stored carbon: **increase** I (leave residues, add roots, perennials) or **decrease** k (waterlogging, cold, no-till, mineral protection). Peat bogs work by the second lever:

waterlogged, oxygen-poor conditions make microbial decay very slow, so partly decayed biomass piles up instead of vanishing. The pool also needs roughly 3τ to reach 95% of C_{eq}, so soil sinks **saturate**.

Worked example

Forest soil: $I = 2$ tC/ha/yr, $k = 0.05\,\text{yr}^{-1} \Rightarrow C_{\text{eq}} = 40$ tC/ha, $\tau = 20$ yr (95% reached in ≈ 60 yr).
Same land as cropland where harvest removes 40% of the input: $I = 1.2 \Rightarrow C_{\text{eq}} = 24$ tC/ha, i.e. 40% *lower*. This reproduces the course's observation that cropland soils hold roughly 30–40% less organic carbon than natural vegetation (Fig. 8).
Peat bog: $I = 1$, $k = 0.01 \Rightarrow C_{\text{eq}} = 100$ tC/ha but $\tau = 100$ yr, so it fills slowly, which is why *protecting existing* peat beats creating new.

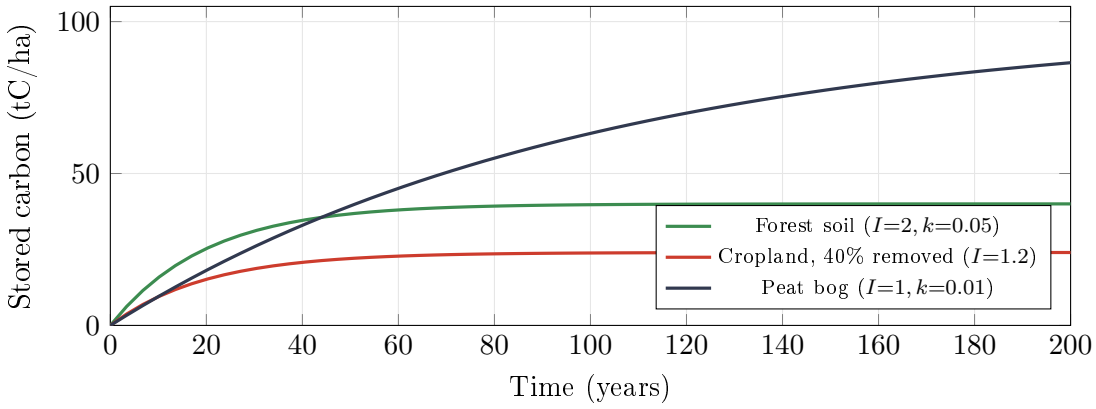


Fig. 8. Same equation, three parameter sets. Each curve levels off, so a soil or forest sink is large but finite. The peat curve is slow but ends highest.

7.2 Forestry practices, defined by what was there before

Fixing a common misconception

Afforestation, reforestation and proforestation sound identical but answer different questions: *was there forest before, and is it healthy?*

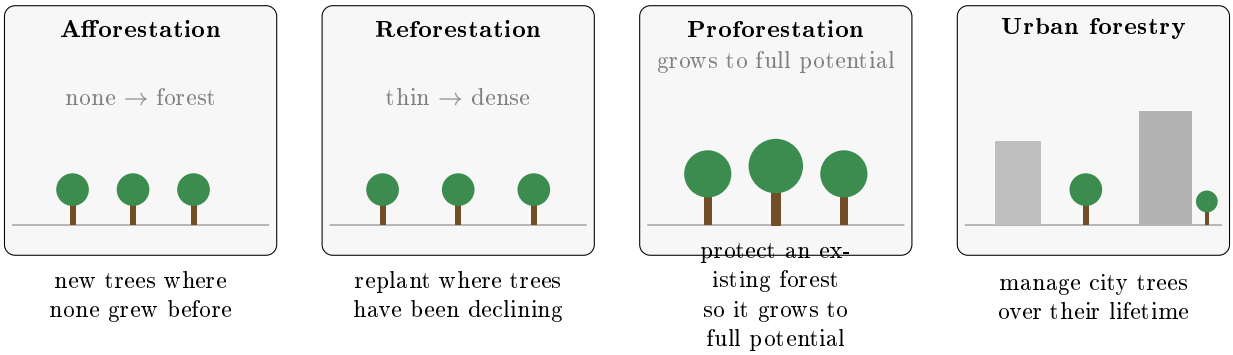


Fig. 9. Four forestry practices in one glance.

Wetlands and blue carbon. Mangroves, seagrass and salt marshes hold about 20–30% of the world's soil carbon (course figure) because waterlogged sediments have very small k (same model).

Agriculture and deep soil. Leaving harvest residues and rotating perennial crops raises I; no-till reduces k and also burns less fuel per acre. Soils hold roughly four times the carbon in the atmosphere, about half of it deep and stabilised by mineral-organic bonds.

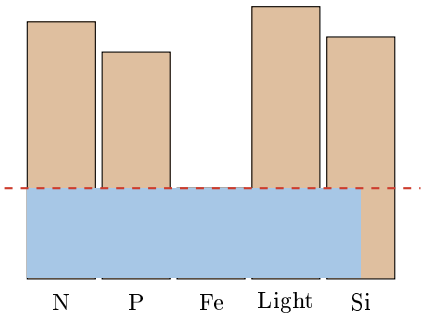
Worked example

Converting carbon mass to CO_2 mass. Photosynthesis, $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, fixes one carbon atom per CO_2. Atomic masses: C = 12, CO_2 = 44, so 1 tonne C $\leftrightarrow$ 44/12 = 3.67 tonne CO_2.
A plantation locking 5 tC/ha/yr for 20 yr on 10 ha stores $5 \times 20 \times 10 = 1000$ tC = 3,667 tCO_2 removed from the air (illustrative figures).

8 Oceanic biological sequestration

WHY this is true

Plankton photosynthesise, take up CO_2, and some sink to deep water when they die, taking carbon with them ("biological pump"). Growth is limited by the scarcest nutrient (Liebig's law of the minimum). In large ocean regions the limiting nutrient is **iron**, so adding iron (or urea, a nitrogen source) should raise plankton growth and carbon export. Sperm whales do this naturally: iron-rich faeces released near the surface fertilise phytoplankton. Artificial fertilisation is controversial because ocean effects are poorly understood (for example nitrogen oxide release and nutrient disruption).



Dashed line = growth. It is set by the **shortest stave** (iron), not by the tall ones. Adding Fe raises it until the next-shortest stave limits.

Fig. 10. Liebig's barrel. Fertilising with something that is not *limiting does nothing, which is why only iron-poor regions respond to iron.*

Mixing layers and seaweed. Vertical pipes can pump nutrient-rich deep water up to trigger algal blooms (similar logic to iron fertilisation). Seaweed grows fast in shallow areas, captures carbon, and can feed biomethane production; the course estimate says 9% ocean coverage could remove about 53 Gt CO_2 per year (a modelling estimate, not an achieved figure).

9 Physical sequestration

9.1 BECCS: why "carbon-neutral" differs from "carbon-negative"

Definition

BECCS (Bioenergy with Carbon Capture and Storage): grow biomass (removes CO_2), convert it to energy (heat, electricity, biofuel via combustion, fermentation or pyrolysis), *capture* the CO_2 and *store* it underground.

Worked example

Ledger in units of "100 CO_2":
Fossil plant: emits +100. With 90% capture: +10.
Biomass plant: growth −100, burning +100, net 0 (**neutral**).
BECCS: growth −100, burning +100, capture −90, net **−90** (**negative**): it removes carbon from the atmosphere while producing energy.

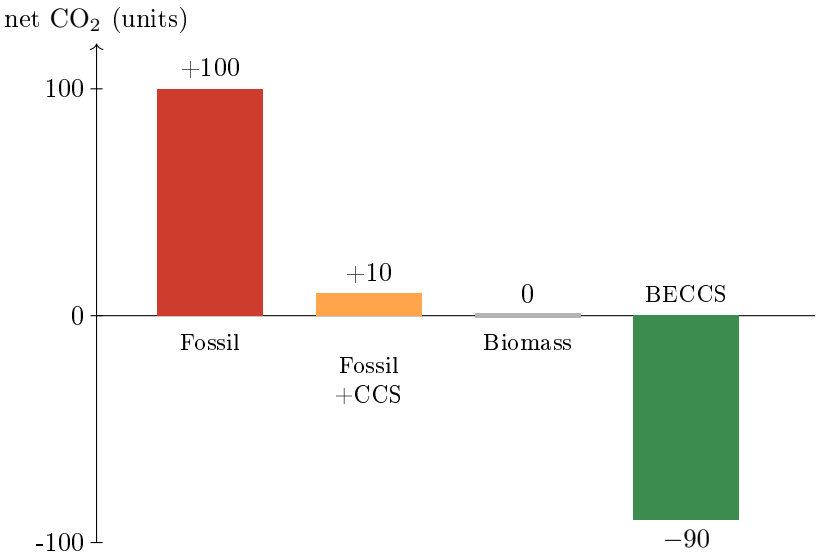


Fig. 11. Only the bar below zero actually reduces atmospheric CO_2. Week 6 revisits why "biomass is carbon-neutral" needs conditions (regrowth time, supply-chain emissions).

9.2 Burial and biochar

Burying wood mimics how fossil fuels formed. **Biochar** is charcoal made by pyrolysis (heating biomass without oxygen); mixed into soil or landfill, a large fraction of its carbon resists decay for very long periods, and it improves soil (terra preta).

9.3 Ocean physical storage

WHY this is true

Gas solubility follows Henry's law: dissolved concentration $= k_H\, p_{CO_2}$. It rises with **pressure** and falls with **temperature** (the course's "directly proportional to pressure, inversely to temperature"). Using $k_H(25°\text{C}) = 0.033$ and a van't Hoff correction, $k_H(5°\text{C}) \approx 0.060$ mol $\text{L}^{-1}\text{atm}^{-1}$. At 420 ppm: dissolved $CO_2 = 1.4\times10^{-5}$ M at 25°C versus 2.5×10^{-5} M at 5°C, i.e. $\approx 1.8\times$ more in cold water. Deep water is both cold and under enormous pressure, hence the proposal to store CO_2 there. Below roughly 3000 m, compressed liquid CO_2 becomes denser than seawater and sinks, forming pools.

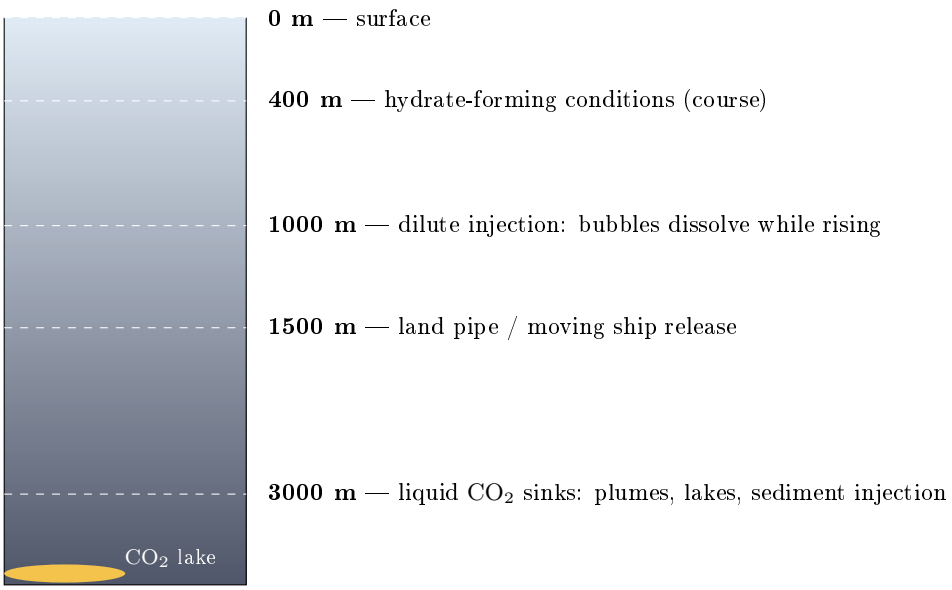


Fig. 12. Depth markers used in the course notes. Deeper means colder, denser and slower mixing, so longer residence time but harder to monitor.

Environmental concerns. Deep-sea life has evolved with very stable chemistry and slow reproduction, so a lethal CO_2 pulse recovers slowly. Extra CO_2 also lowers pH, and CO_2 itself disrupts physiology; long-term effects on surface waters remain unknown.

9.4 Geological storage: why about 1 km down?

WHY this is true

CO_2 becomes a dense **supercritical fluid** above its critical point: $T_c = 31.1°C$ and $P_c = 73.8$ bar. Depth supplies both conditions.
Pressure (hydrostatic): $P \approx 1 + 0.098\,h$ bar (h in m). $P = 73.8$ bar at $h = (73.8-1)/0.098 \approx 740$ m.
Temperature (geothermal gradient, 1°C per ~ 37 m from the geothermal notes): $T \approx 15 + h/37$. At 740 m, $T \approx 35°C > 31.1°C$.
So below roughly 750 m both conditions hold; ~ 1 km gives a margin: $P \approx 99$ bar, $T \approx 42°C$.

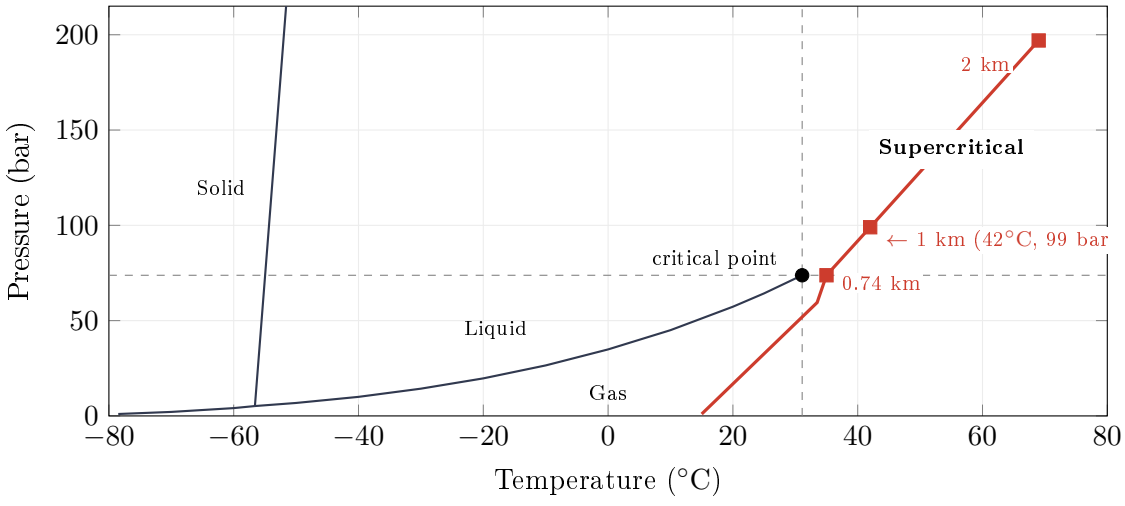


Fig. 13. Schematic CO_2 phase diagram (boundaries drawn from standard data points). The red line is the pressure–temperature you meet going downward. It crosses into the supercritical region at about 0.74 km and stays inside.

Worked example

Why compress it? At the surface (15°C, 1 atm) CO_2 has density ≈ 1.87 kg/m^3, so 1 t occupies ≈ 535 m^3. At 1 km depth the supercritical density is ≈ 630 kg/m^3 (typical value), so 1 t occupies ≈ 1.6 m^3, about 340× less volume. Storage in pores of a deep saline formation, depleted oil/gas reservoir or unminable coal bed becomes feasible; stability is expected over millions of years provided the caprock holds.

10 Chemical sequestration: mineral carbonation

WHY this is true

Calcium and magnesium oxides and silicates react with CO_2 to form stable carbonates, which sit at a much lower energy than gaseous CO_2 (the same reaction as rock weathering, but speeded up):

$$CaO + CO_2 \to CaCO_3, \qquad MgO + CO_2 \to MgCO_3, \qquad Mg_2SiO_4 + 2CO_2 \to 2MgCO_3 + SiO_2.$$

Mass balance (atomic masses Ca 40, Mg 24.3, Si 28, O 16, C 12): 1 t CaO (56 g/mol) binds $44/56 = 0.78$ t CO_2; 1 t forsterite (Mg_2SiO_4, 140.7 g/mol) binds $88/140.7 = 0.63$ t CO_2. A large tonnage of rock is therefore needed, which is the main practical cost.

Method	Where carbon goes	Permanence	Main limit or risk
Forests, soils, wetlands	plants and soil	decades–centuries; saturates ($\sim 3/k$)	fire, drainage, land use change
Peat	waterlogged soil	centuries if kept wet	drainage releases it fast
Ocean iron/urea	deep ocean via plankton	uncertain	ecosystem side effects, poorly understood
Seaweed farming	biomass, biomethane	depends on use	scale, ocean impacts
BECCS	underground	very long (if stored)	needs land for biomass plus CCS infrastructure
Biochar and burial	soil or landfill	long	feedstock supply
Deep ocean injection	deep water or sediment	centuries	deep-sea ecosystems, pH
Geological storage	rock pores at $\sim$ 1 km	up to millions of years	leakage, site suitability
Mineral carbonation	solid carbonate	geological	large rock tonnage, energy

11 Environmental health monitoring with sensors, IoT and AI

Definition

Environmental health monitoring is the systematic collection, analysis and assessment of environmental data (air, water, soil, biodiversity) to detect change, find health risks, track trends and predict future conditions, so that authorities can act. Its end goal is protecting public health, not merely recording numbers.

Fixing a common misconception

"Monitoring reduces pollution." Monitoring only *measures*. Pollution falls when a reading triggers an **action** (a notice, a shutdown, a warning, a policy). A dashboard that nobody acts on is just a record. Likewise, a high reading is *hazard* data; *risk* also needs how many people are exposed and for how long.

11.1 The pipeline every system shares

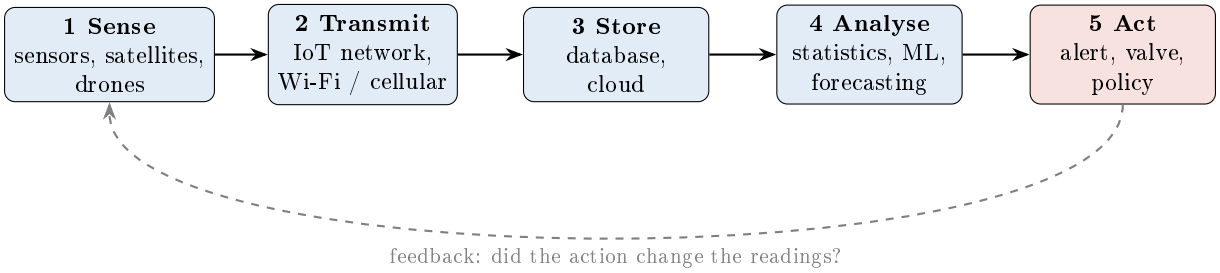


Fig. 14. Five stages. The dashed loop is what separates a monitoring system *from a one-off survey.*

11.2 How often must you measure? A sampling argument

WHY this is true

An event of duration D (a pollution spike, an effluent release) can be missed completely if samples are spaced further apart than D. A safe rule of thumb is to sample at least about *twice per event duration*, i.e. interval $\leq D/2$ (the same idea as Nyquist sampling). Too slow and you miss events; too fast and you waste power, bandwidth and storage, which matters for battery-powered remote sensors.

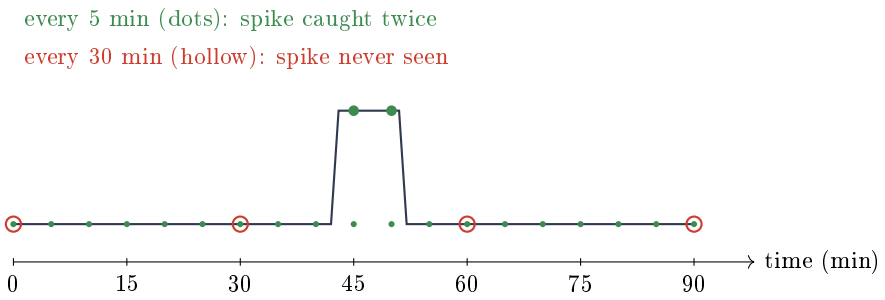


Fig. 15. A 10-minute spike (here, minutes 42–52) passes between the 30-minute samples.

Worked example

Filtering noise without hiding real events. Turbidity readings (NTU) arrive as 4.1, 4.3, 9.8, 4.2, 4.4 (one stray spike) and later 6.0, 6.8, 7.4 (a sustained rise). Rule: raise an alert when the **3-point average** exceeds 6.5 (illustrative threshold).
Spike case: $(4.3 + 9.8 + 4.2)/3 = 6.1 < 6.5 \Rightarrow$ no alert (likely noise).
Sustained case: $(6.0 + 6.8 + 7.4)/3 = 6.73 > 6.5 \Rightarrow$ alert.
Trade-off: averaging reduces **false alarms** but delays **detection**.

11.3 What gets measured, and with what

Medium	Parameters / devices	Why it matters
Air	$PM_{2.5}$, PM_{10}, CO, SO_2, NO_x, VOCs; low-cost sensors, satellites, e-noses	hotspots and early health warnings
Water	pH, dissolved oxygen, turbidity, chemical contaminants; in-water probes	early detection of effluent or contamination
Soil	moisture, temperature, pH, NPK, contamination; soil probes	food safety, irrigation, farmland health
Wildlife	acoustic sensors, camera traps, GPS tags	population tracking, illegal logging or poaching
People	wearable exposure sensors	personal exposure in real time

Why $PM_{2.5}$ gets special attention. The number is particle diameter in micrometres. A human hair is about 70 μm, so a $PM_{2.5}$ particle is roughly 28× narrower: small enough to travel deep into the lungs.

11.4 Remote sensing, with a computed example (NDVI)

Definition

Remote sensing gathers information about an object or area without touching it, usually from satellites, aircraft or drones that record reflected or emitted radiation. (The course abbreviation "RMS" appears to be a slip for RS.) Applications: agriculture, forestry, land cover, water resources.

WHY this is true

Different surfaces reflect different wavelengths. Healthy leaves **absorb red light** (chlorophyll) but **strongly reflect near-infrared (NIR)** (leaf cell structure). Water absorbs both. The Normalised Difference Vegetation Index uses that contrast:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} \in [-1, 1].$$

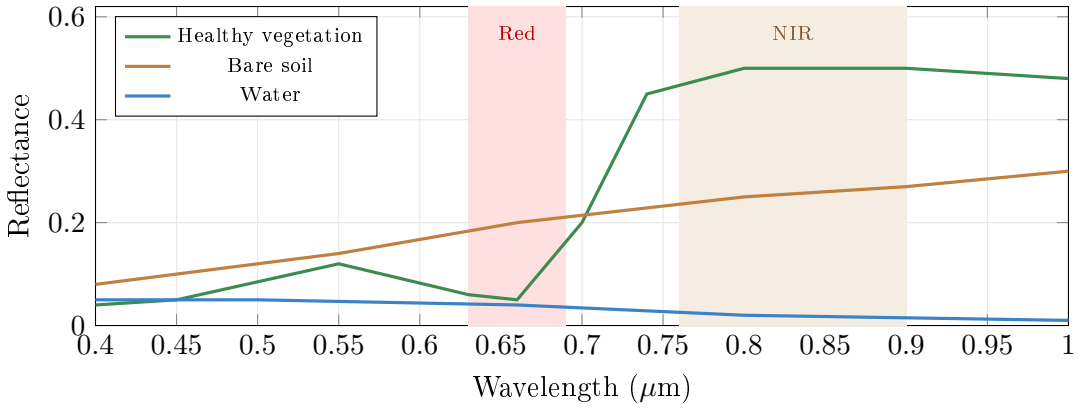


Fig. 16. Schematic reflectance curves (illustrative values). Vegetation's "red edge" near 0.7 μm is why NIR–Red contrast isolates plants.

Worked example

Vegetation: NIR = 0.50, Red = 0.08 ⇒ $(0.50 - 0.08)/(0.58) = \mathbf{0.72}$.
Bare soil: NIR = 0.25, Red = 0.20 ⇒ $0.05/0.45 = \mathbf{0.11}$.
Water: NIR = 0.02, Red = 0.04 ⇒ $-0.02/0.06 = \mathbf{-0.33}$.
High positive = dense healthy plants; near zero = bare ground; negative = water. A forest cleared by illegal logging drops from ∼ 0.7 toward ∼ 0.1 in one image comparison.

11.5 Databases and information systems

A **database** is a collection of inter-related data on various subjects. The course lists: the Ministry of Environment and Forests; **NMIS** (National Management Information System: research and development projects with scientists and personnel); and **ENVIS** (Environmental Information System, a network of database centres).

Check against your course notes

The course sentence reads "ENVIS functions in 25 countries all over the country": that is almost certainly a typo for centres/places across India. Learn it as an Indian environmental-information *network*, and follow the course wording if an MCQ uses it.

11.6 Where AI genuinely helps, and where you must be careful

- **Forecasting:** sequence models such as LSTM learn how past readings predict the next ones (air quality, floods, CO_2, sea level).
- **Detection:** deep learning on satellite/aerial images maps habitat, deforestation, illegal dumping; anomaly detection flags unusual boat tracks or sudden sensor changes.
- **Simulation:** digital twins (virtual copies of, say, a treatment plant) test operating choices safely; e-noses recognise gases from chemical signatures; robots enter hazardous sites.

Fixing a common misconception

"An AI model that predicts air quality must beat simple methods." Not automatically. Always compare against baselines.

Worked example

Hourly $PM_{2.5}$ ($\mu g/m^3$): 38, 42, 45, 41, 47, 52. Predict hours 4, 5, 6.
Persistence (next = last): forecasts $45, 41, 47$; absolute errors $4, 6, 5$; MAE = 5.0.
3-point moving average: forecasts 41.67, 42.67, 44.33; errors 0.67, 4.33, 7.67; MAE = 4.22.
A slightly better rule from two lines of arithmetic; but both lag behind the rising trend, which is exactly what a trained model must learn to beat. Report an AI model's error *next to* these baselines.

11.7 Two case studies from the course

	Problem	AI approach	Impact
IIT Kharagpur, arsenic (Ganga delta)	arsenic in groundwater drinking sources across eastern India	AI model analyses geo-science data to separate high- and low-arsenic zones and quantify exposure	identifies safe groundwater for public-health action
The Ocean Cleanup	plastic pollution in remote oceans	algorithms detect debris and map litter	targeted, more efficient clean-up

12 Week 5 summary tables and self-test

Idea	Core relation or fact	Remember
Surface temperature without greenhouse gases	$T_e = [S_0(1-\alpha)/4\sigma]^{1/4} = 255$ K	+33 K from natural effect
One-layer model	$T_s = 2^{1/4} T_e \approx 303$ K	240 in, 480 at surface, 240 out
Wien's law	$\lambda_{\max} = 2898/T$ μm	Sun 0.5 μm, Earth 10 μm
GWP	per kg versus CO_2	fluorinated gases up to $\sim$ 23,000
Feedback	positive = amplifying	ice–albedo, permafrost
Water vapour	$+\sim 7\%$ per °C	heavier rain and drought
pH	$[H^+] = 10^{-\mathrm{pH}}$	-0.1 pH $\Rightarrow +26\%$ H^+
Carbon mass to CO_2	$\times 44/12 = 3.67$	
Soil/peat pool	$C_{\mathrm{eq}} = I/k$, $\tau = 1/k$	sinks saturate
Supercritical CO_2	$T > 31.1$°C, $P > 73.8$ bar	$\approx$ 0.75–1 km depth
BECCS	capture > burn emissions $\Rightarrow$ net negative	neutral $\neq$ negative
NDVI	$(\mathrm{NIR} - \mathrm{Red})/(\mathrm{NIR} + \mathrm{Red})$	plants ≈ 0.7

Course numbers to memorise (these are what MCQs test)

Fact	Value in the course notes
Snow and ice reflect	about 90% (text); figure shows 85%; ocean absorbs 93%; about 70% of sunlight reaching Earth is absorbed
Sea level since 1880	about 8 inches, 1–2 mm/yr; by 2100 about 1–8 feet higher
Arctic warming	about twice as fast as elsewhere
Maldives	1,192 small islands, about 80% less than 1 m above sea level
Human cause of warming since 1950s	extremely likely ($> 95\%$)
Record CO_2	414.8 ppm; 52% above pre-industrial (2020)
Fluorinated gases	up to 23,000× CO_2
Permafrost	frozen for up to 40,000 years
Vertebrate extinction (2015 study)	114× faster than normal
Ocean share of human CO_2 absorbed	more than 25%
Wetlands	20–30% of world soil carbon
Cropland soil carbon	30–40% less than natural vegetation
Soil carbon	4× the atmosphere; more than vegetation plus atmosphere
Seaweed farms on 9% of ocean	biomethane for world fossil demand, 53 Gt CO_2/yr removed, 200 kg fish/person/yr for 10 billion people
Geological storage	about 1 km depth, supercritical fluid, stable for up to millions of years
Ocean injection depths	dilute CO_2 at 1000 m; hydrate/solid sinks (1.5× heavier than seawater); deep sediments at least 3000 m; plumes at 3 km; ocean could hold over 1000 billion tonnes
Monitoring case studies	IIT Kharagpur (arsenic, Ganga delta); The Ocean Cleanup (plastic)

Self-test checklist (try each without looking)

- ☐ Compute Earth's no-atmosphere temperature and explain what the 33 K gap represents.
- ☐ Explain why sunlight passes through the atmosphere but infrared does not, using wavelengths.
- ☐ Show that a one-layer atmosphere doubles the surface radiation, and check the balance at each level.
- ☐ Convert 1 t of CH_4 to CO_2-equivalent and explain why GWP is per kg, not a total.
- ☐ Name three natural drivers and say why each fails to explain warming since 1950.
- ☐ Draw the ice–albedo loop and compute the six-fold absorption difference.
- ☐ Compute the percent rise in $[H^+]$ for a 0.1 pH drop, and say why the ocean is not "acidic".
- ☐ Explain why melting sea ice barely changes sea level but melting land ice does.
- ☐ Use $C_{\text{eq}} = I/k$ to explain peat, cropland carbon loss and why sinks saturate.
- ☐ Derive why CO_2 is stored at about 1 km (critical point + hydrostatic pressure + geothermal gradient).
- ☐ Do the BECCS ledger and state the difference between neutral and negative.
- ☐ Compute NDVI from three reflectances and say what each value means.
- ☐ Describe the sensing–acting pipeline and what sampling interval catches a 10-minute event.